

Search for Radio Counterparts associated with Fast Radio Bursts



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Introduction

Fast radio bursts (FRBs) are millisecond-duration radio pulses from outside the Milky Way whose physical origins remain unknown¹. Studying their **local environments** can provide important clues about their progenitors.

CHIME/FRB, the fast radio burst project of the Canadian Hydrogen Intensity Mapping Experiment, is the world's most prolific FRB detector, having discovered over 4,500 bursts to date².

Persistent radio sources (PRSs) are compact radio sources located at or near the position of an FRB. Unlike the millisecond-duration FRB, PRSs remain detectable over much longer timescales, making them **useful probes of FRB local environments**.

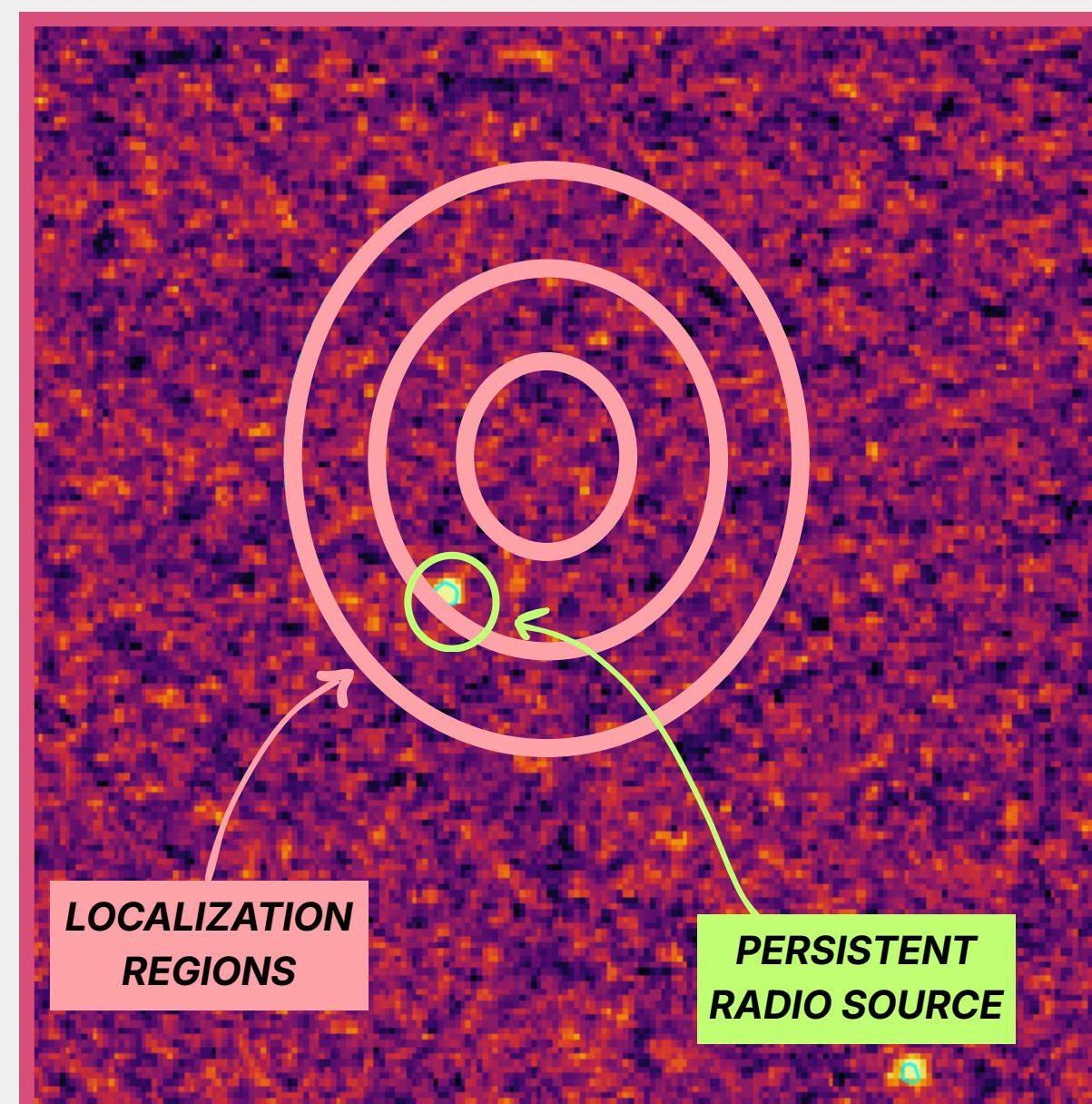
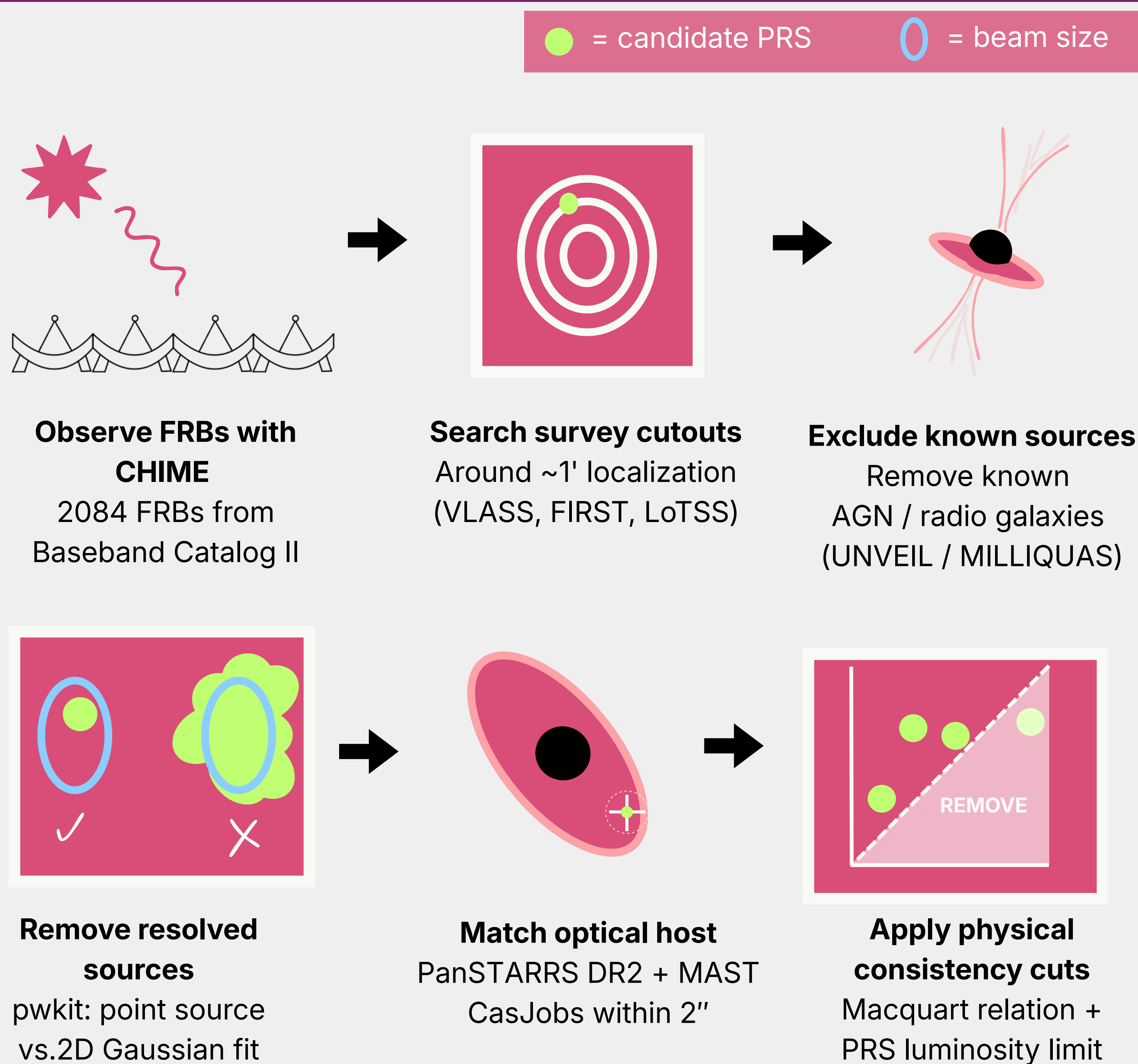


Figure 1: VLASS image of FRB20210923B, showing the FRB localization region (pink) & a coincident PRS (green).

Previous searches of 37 CHIME/FRB repeaters identified two candidate PRSs³, one of which was later confirmed to be spatially coincident with FRB 20190417A⁴. **We extend this approach to 2084 FRBs, including both repeaters and non-repeaters.**

Methods



Conclusion

- Eight candidate PRSs** remain after radio, optical, DM vs. redshift, and luminosity selection cuts.
- Further uncertainty analyses needed to refine the sample and **rule out AGN/radio galaxy contamination**, followed by higher resolution and multi-wavelength observations for confirmation.
- The sample includes candidates associated with non-repeating FRBs; confirming them will **help determine whether PRSs occur across the broader FRB population or are preferentially associated with repeaters.**

Results

After all selection cuts, we find 8 candidate persistent radio sources.

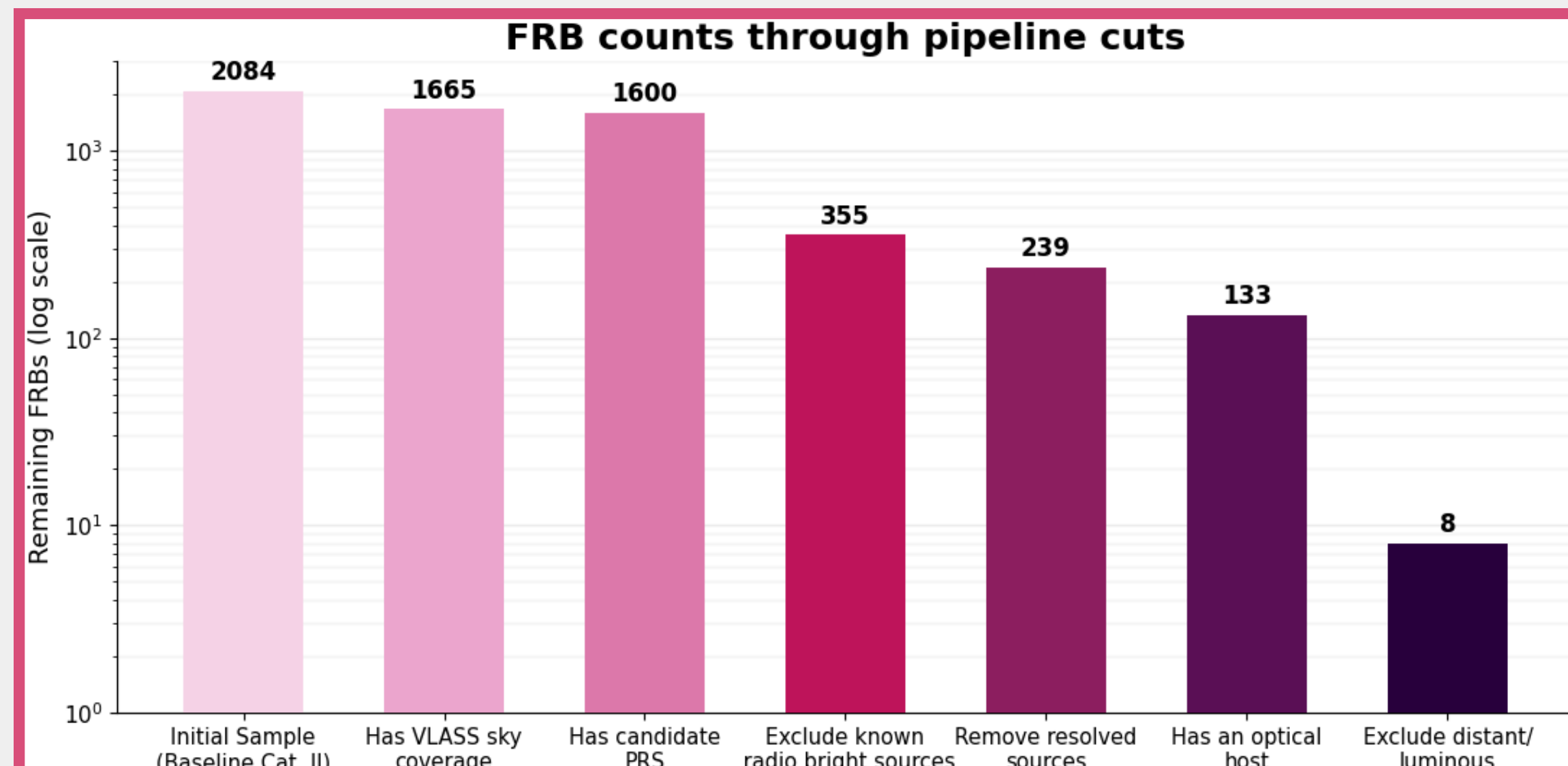


Figure 2: Number of FRBs remaining after each stage of the PRS candidate selection pipeline, reducing the initial sample of 2084 FRBs to 8 FRBs with one candidate each.

The final 8 candidates are consistent with the expected FRB dispersion measure (DM) - redshift relation & compared with the L_v - rotation measure (RM) properties of previously known PRSs.

This relation predicts that DM grows with redshift as intergalactic electron density builds along the sightline.

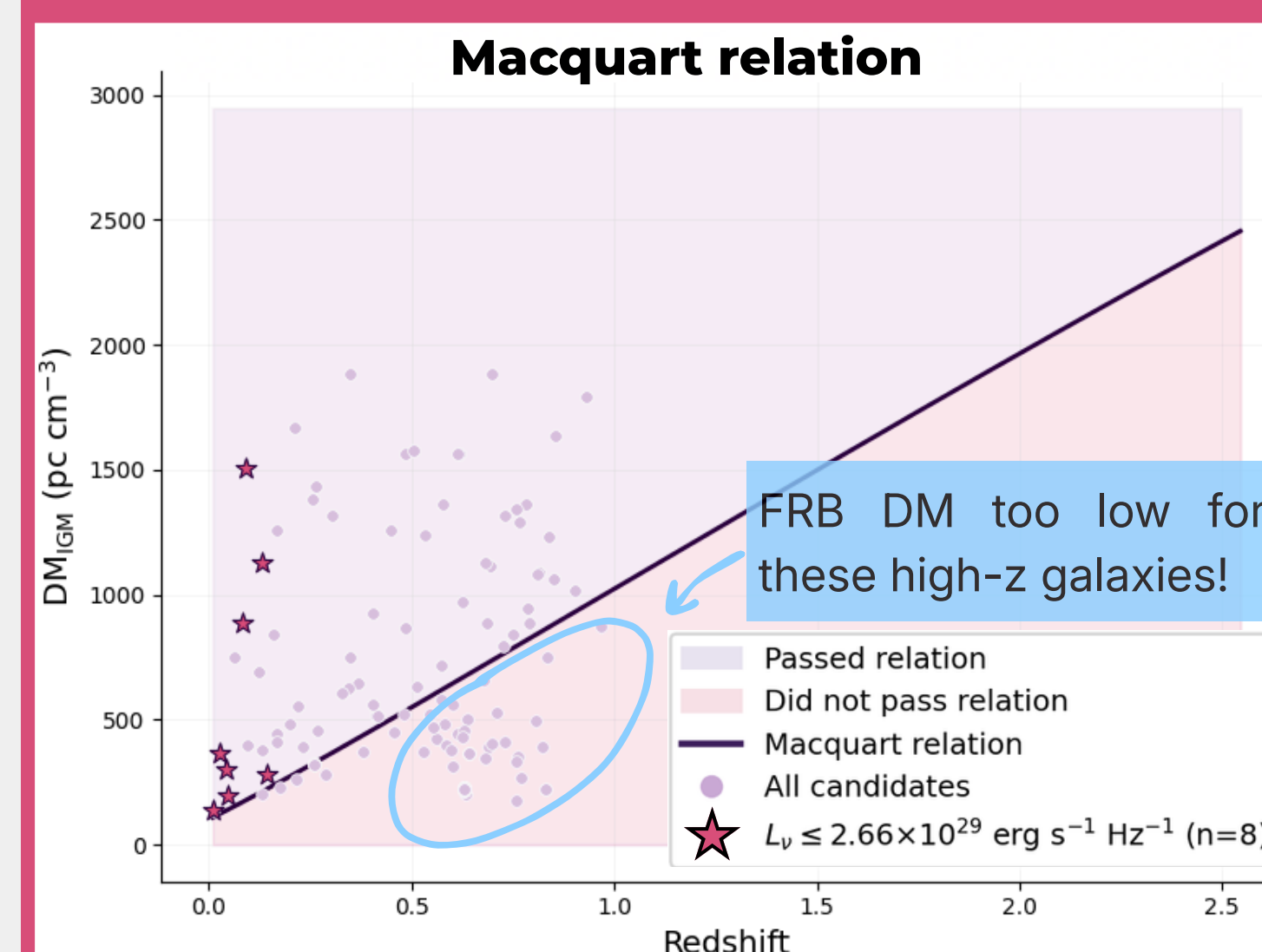


Figure 3: Dispersion measure vs. redshift for the candidate sample. Final PRS candidates (starred) remain consistent with the Macquart relation.

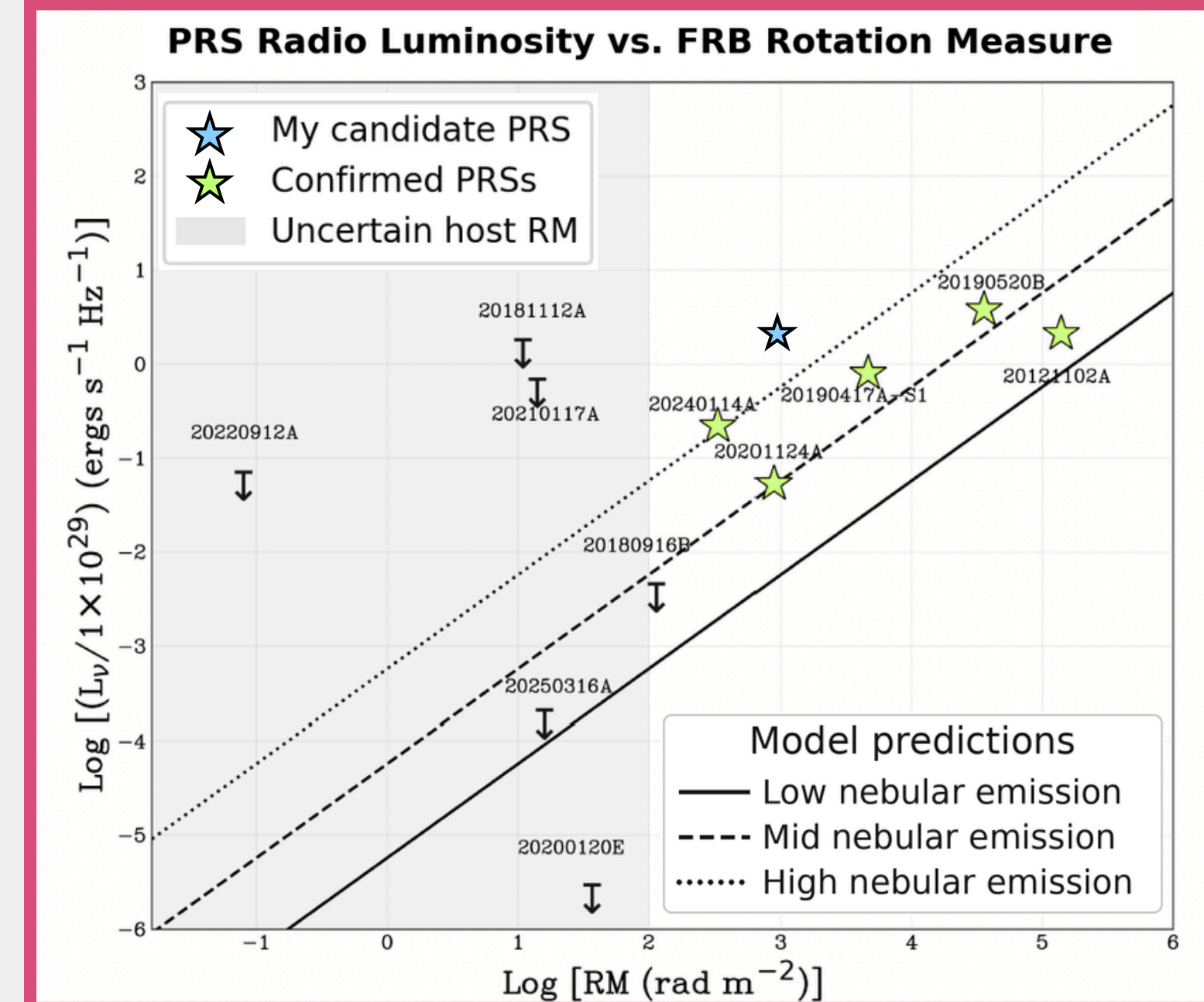


Figure 4: PRS radio luminosity vs. FRB rotation measure, with predictions from the nebulae PRS model (Bruni et al. 2024). My candidate PRS falls near the confirmed PRS population, consistent with the predicted relation.

Two particularly promising candidates include:

FRB2021032BC: This FRB has a high RM ($RM = 974.10 \text{ rad m}^{-2}$), and the candidate PRS has a luminosity broadly consistent with the predicted L_v - RM relation (Figure 4).

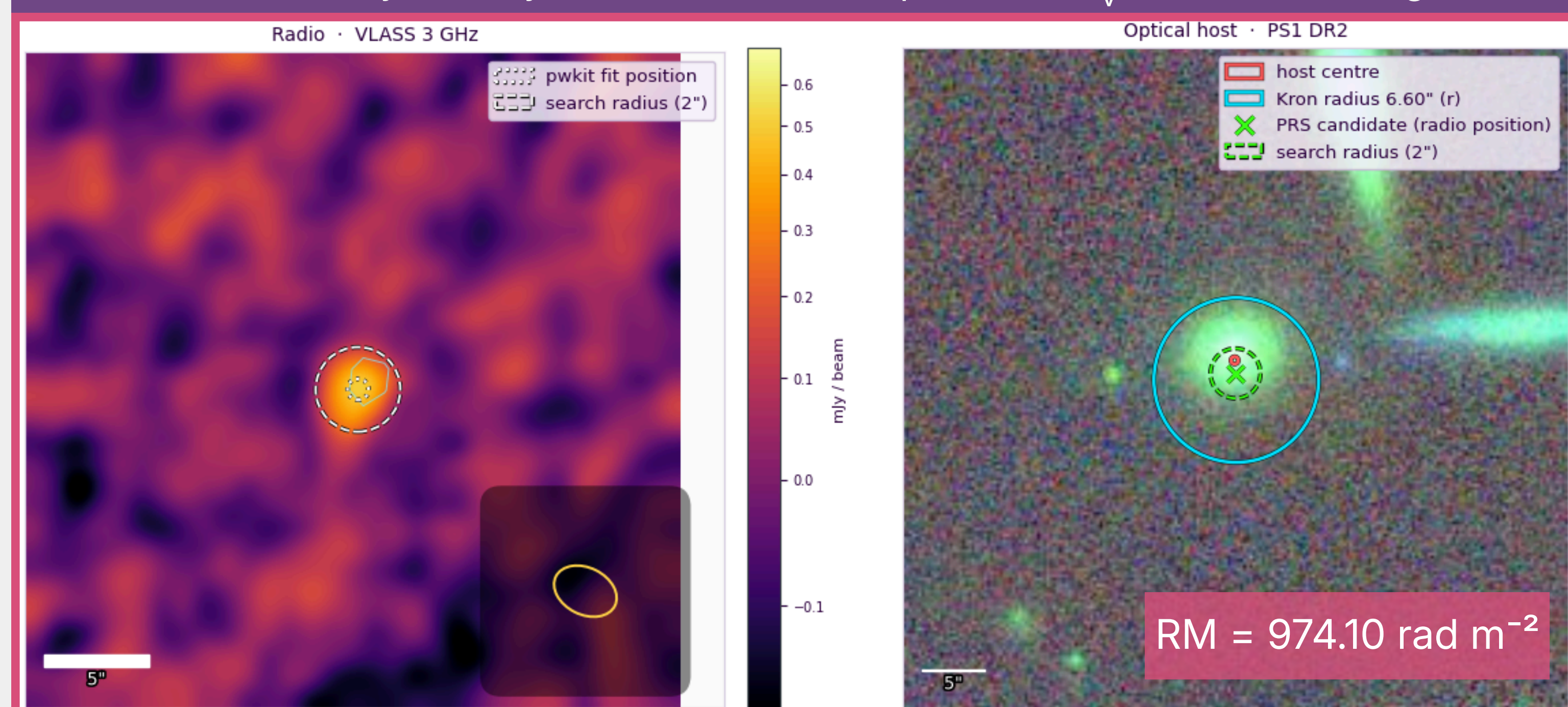


Figure 5: VLASS radio and Pan-STARRS optical images of FRB2021032BC.

FRB20230411D: This candidate PRS is clearly offset from the centre of its putative host galaxy, making it unlikely to be either an AGN or knot of star formation.

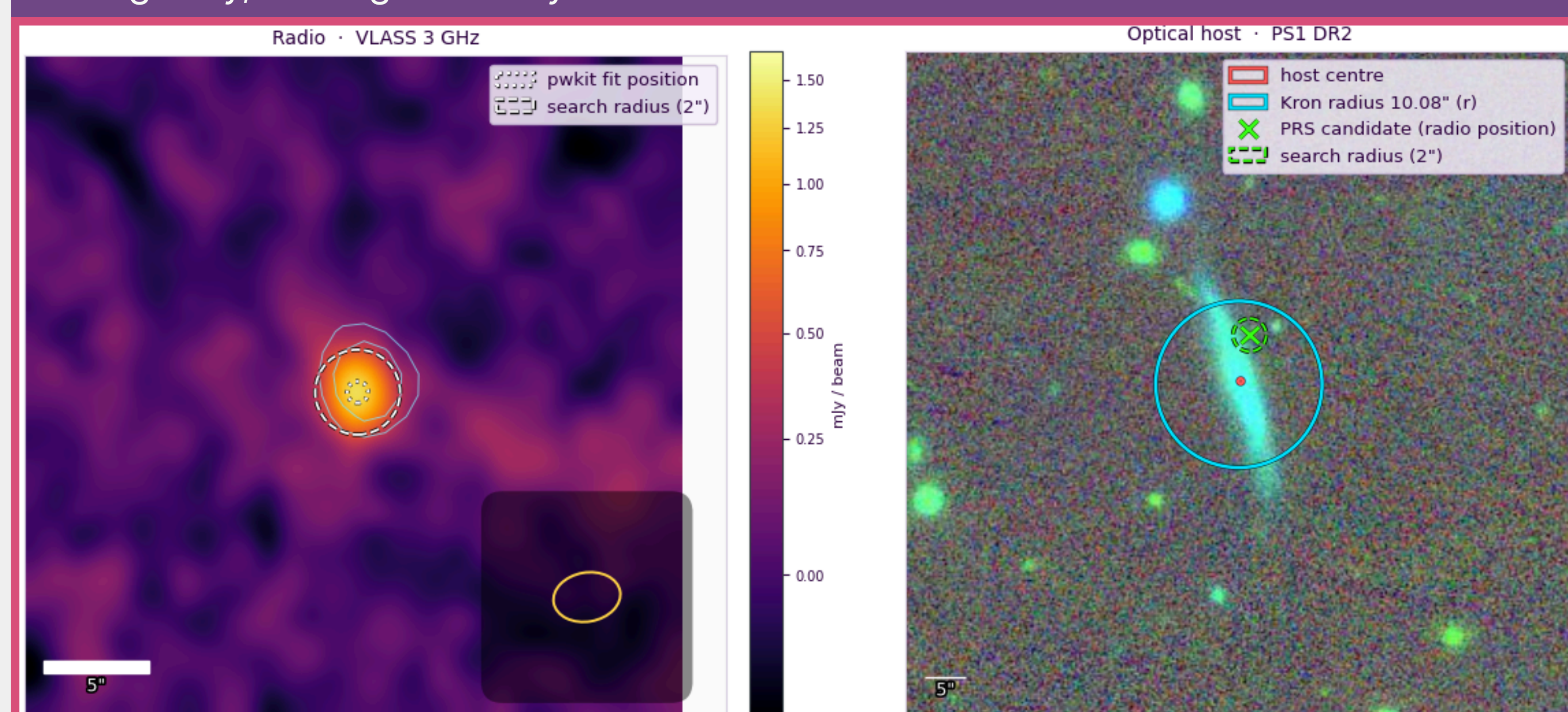


Figure 6: VLASS radio and Pan-STARRS optical images of FRB20230411D.